

From High-School Maths to HATI

A self-contained course in the mathematics of
lunar terrain-hazard detection

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For anyone who can solve a quadratic and wants to read this and mean it:

$$H(x) = \sigma \left(b + \frac{1}{\|\mathbf{w}\|_1} \sum_{k=1}^{10} w_k \operatorname{clip}_{\pm c} \left[\frac{g(C_k[z])(x) - \mathbf{med} \, g(C_k[z])}{\operatorname{IQR} \, g(C_k[z])/1.349} \right] \right).$$

By the last page this will not look like noise. It will look like a sentence.

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Start here

The equation on the page before this one is the whole of HATI’s elevation pipeline. Right now it is probably a wall of symbols. That is fine. The job of this book is to take you, one idea at a time, from ordinary high-school maths to the point where you can read that line aloud and explain every piece of it to someone else.

HATI is a tool that looks at a map of the Moon and decides where a lander should not touch down. It does that with two kinds of mathematics: the geometry and calculus of a bumpy surface, and the statistics and machine learning that turn “bumpy” into a single number between 0 and 1. Both are within your reach. Neither requires anything you could not have met by the end of school; it just requires that we build the ideas in the right order and never skip the meaning.

Intuition first

Think of the master equation as a recipe written in shorthand. A chef reading “fold in the egg whites” sees a whole technique in four words. By the end of this book each clump of symbols will be like that for you: a technique you recognise on sight. We are going to learn the techniques first, then come back and read the recipe.

An honest word. Reading this book will give you genuine understanding. It will not, by itself, make you an expert. Nobody became fluent in a language by reading a grammar. Expertise comes from the PRACTICE boxes — do them, with a pen, before looking at the solutions in Appendix B. The maths only becomes yours when your own hand has produced it.

What you need before page one. Comfort with: algebra (rearranging $y = mx + c$), basic functions and graphs, right-angled trigonometry ($\sin$, $\cos$, $\tan$), and the idea of a square root and a power. That is all. Everything past that — the calculus, the statistics, the linear algebra, the machine learning — we build here.

How the book is built. Six parts. Parts I–III build the language and the tools. Part IV is the machine learning. Part V is the geometry and signal theory. Part VI returns to the master equation and reads it, symbol by symbol. Every chapter ends with a purple box, *Where this lives in HATI*, tying the idea back to the real pipeline, so you never learn anything without knowing why.

Part I

The language of the machine

Chapter 1

Functions, graphs, and three special curves

A *function* is a machine: a number goes in, a number comes out, and the same input always gives the same output. We write $f(x)$ for “the output when the input is x .” Three particular functions run through all of HATI, so we meet them now.

1.1 The straight line, and what a weighted sum is

The line $f(x) = mx + c$ you already know: m is the slope (how fast output climbs as input grows), c the starting height. Now stretch the idea. Suppose the output depends on *several* inputs $x_1, \dots, x_n$, each with its own importance w_k :

$$f(x_1, \dots, x_n) = w_1x_1 + w_2x_2 + \dots + w_nx_n + b = \sum_{k=1}^n w_kx_k + b. \quad (1.1)$$

That big- Σ is just “add these up”; $\sum_{k=1}^n a_k$ means $a_1 + a_2 + \dots + a_n$. A *weighted sum* is the single most important shape in this whole book: each input gets multiplied by how much it matters, and the results are added. Hold onto it.

1.2 The exponential and the logarithm

The exponential e^x is the function that grows in proportion to how big it already is (interest on interest). Its number $e \approx 2.718$ is just the natural base for that kind of growth. Two facts we will use:

$$e^0 = 1, \quad e^x \text{ is always positive}, \quad e^x \rightarrow 0 \text{ as } x \rightarrow -\infty. \quad (1.2)$$

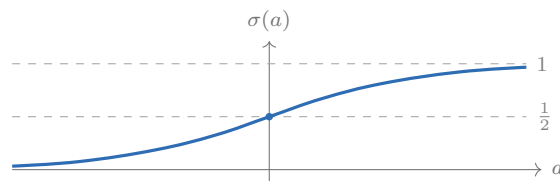
The logarithm $\log$ is its exact reverse: $\log(e^x) = x$. It turns multiplication into addition and, for us, it *tames* numbers with a huge range, squashing a long tail down to something manageable. We will use $\log(1+x)$ specifically, because it behaves at $x=0$ (giving 0, not $-\infty$).

1.3 The logistic curve: the S that makes a decision

Now the star. Define

$$\sigma(a) = \frac{1}{1 + e^{-a}} \quad (\text{the logistic or sigmoid}). \quad (1.3)$$

Feed it any number on the line, $-\infty$ to $+\infty$, and it answers with a number strictly between 0 and 1. Check the ends: if a is very negative, e^{-a} is huge, so $\sigma \rightarrow 0$; if a is very positive, $e^{-a} \rightarrow 0$, so $\sigma \rightarrow 1$; and $\sigma(0) = \frac{1}{2}$. It is a smooth switch.



(The curve is drawn with the output axis stretched so the flat-to-1 part is visible; the shape is the point.)

The key idea

A weighted sum produces a single number a that can be anything. The sigmoid $\sigma(a)$ then bends that number into a probability between 0 and 1. “Weighted sum, then sigmoid” is the entire skeleton of HATI’s decision — and, as we will see in Part IV, of the simplest machine-learning classifier ever built.

Where this lives in HATI

In the master equation, the outermost symbol is $\sigma(\dots)$: the very last thing HATI does is squash a weighted sum through this S-curve to get the hazard probability $H(x)$. The $g = \log(1 + \cdot)$ inside it is the tail-taming log you just met. You already recognise two pieces of the wall.

Practice

1.1 Compute $\sigma(0)$, and argue without a calculator whether $\sigma(2)$ is closer to 0 or to 1. **1.2** A weighted sum has weights $w = (2, -1)$, bias $b = -1$, inputs $x = (1.5, 1)$. Find $a = \sum w_k x_k + b$, then $\sigma(a)$ (a calculator is fine). **1.3** Why would $\log(x)$ be a bad “tail-tamer” for a quantity that can equal 0, and why does $\log(1 + x)$ fix it?

Chapter 2

The trigonometry you will actually use

You met sin, cos, tan as ratios of sides in a right triangle. HATI uses exactly two ideas from that: turning a ratio into an angle, and turning an angle into a length.

2.1 Slope is an angle

Walk one metre east and rise r metres. The steepness is the ratio $r/1 = r$ (“rise over run”), but it is often nicer as an *angle* θ above horizontal, and the two are linked by the tangent:

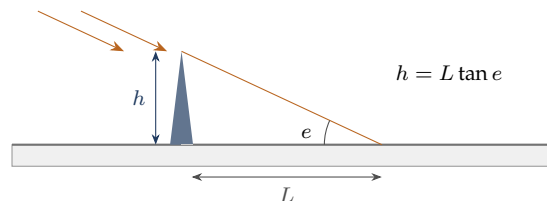
$$\tan \theta = \frac{\text{rise}}{\text{run}} \iff \theta = \arctan\left(\frac{\text{rise}}{\text{run}}\right). \quad (2.1)$$

$\arctan$ (“inverse tangent”) is just the button that turns a ratio back into the angle that produced it. A flat surface has $\theta = 0$; a 45° slope has $\tan \theta = 1$.

2.2 A shadow is a ruler

Here is the other side of the same triangle. The Sun sits at an angle e above the horizon. An object of height h casts a shadow of length L on flat ground. The Sun ray grazes the top of the object and hits the ground at the shadow’s tip, making a right triangle with the object (height h) and the shadow (length L):

$$\tan e = \frac{h}{L} \implies \boxed{h = L \tan e.} \quad (2.2)$$



The lower the Sun (small e), the longer the shadow for a given height — which is why the lunar pole, where the Sun barely clears the horizon, turns even tiny rocks into long, measurable shadows.

Where this lives in HATI

Channel 2 of HATI measures shadow lengths L in a photograph and uses Equation (2.2) to recover obstacle heights h — seeing rocks far too small for the elevation map. The slope-as-angle idea (Section 2.1) is how HATI turns the steepness of the terrain into the slope field it then measures the roughness of.

Practice

2.1 A boulder casts a 6 m shadow with the Sun 20° up. How tall is it? **2.2** The same boulder at sunset, Sun only 5° up — now how long is its shadow? What does this say about the best time to hunt for small hazards?

Chapter 3

Vectors and the dot product

HATI describes each spot on the Moon by ten numbers at once. To handle “ten numbers as one object” we need vectors.

Definition

A *vector* is an ordered list of numbers, written $\mathbf{v} = (v_1, v_2, \dots, v_n)$. The number n is its *dimension*. A point on a map is a 2-vector (X, Y) ; a spot described by ten roughness numbers is a 10-vector.

3.1 The dot product is a weighted sum in disguise

Given two vectors of the same length, their *dot product* multiplies matching entries and adds:

$$\mathbf{w} \cdot \mathbf{x} = \mathbf{w}^T \mathbf{x} = \sum_{k=1}^n w_k x_k = w_1 x_1 + \dots + w_n x_n. \quad (3.1)$$

Look familiar? It is exactly the weighted sum from Chapter 1. So if $\mathbf{x}$ holds the ten roughness numbers for a spot and $\mathbf{w}$ holds how much each one matters, then $\mathbf{w} \cdot \mathbf{x}$ is “the overall hazard vote” in a single product.

Intuition first

A vector is a labelled bag of numbers you can carry around as one thing. The dot product asks: “how much do these two bags agree, weighted entry by entry?” Big positive dot product means the inputs that matter most are large. That is precisely the score we want to feed the sigmoid.

Where this lives in HATI

The fraction $\frac{1}{\|\mathbf{w}\|_1} \mathbf{w}^T \boldsymbol{\zeta}$ inside the master equation is a dot product: $\boldsymbol{\zeta}$ is the vector of ten standardised roughness numbers, $\mathbf{w}$ the literature weights, and $\|\mathbf{w}\|_1 = \sum_k |w_k|$ is a scaling so the result does not balloon as we add channels. The whole of Chapter 7 in `hati_math.pdf` is built on this one product.

Practice

3.1 With $\mathbf{w} = (1, 0.7, 0.7)$ and $\mathbf{x} = (2, -1, 3)$, compute $\mathbf{w} \cdot \mathbf{x}$. **3.2** If every weight is positive and one input doubles while the rest stay fixed, can the dot product go down? What does that tell you about how a single rough channel affects HATI’s vote?

Part II

Calculus on a grid

Chapter 4

Derivatives and the gradient

4.1 The derivative: instantaneous steepness

Zoom in on the graph of $f(x)$ until the curve looks straight. The slope of that tiny straight piece is the *derivative* $f'(x)$, written also $\frac{df}{dx}$. It is the rate of change: how fast the output moves when you nudge the input. Formally it is the limit of rise-over-run as the run shrinks to nothing,

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}, \quad (4.1)$$

but the picture — “slope of the curve right here” — is what you need.

Worked example

For $f(x) = x^2$, nudging from x to $x + \Delta x$ changes the output by $(x + \Delta x)^2 - x^2 = 2x\Delta x + \Delta x^2$. Divide by Δx : $2x + \Delta x$. As $\Delta x \rightarrow 0$ this is $2x$. So $f'(x) = 2x$: at $x = 3$ the curve rises three-to-one steeper than at $x = 1$. We just did calculus from scratch.

4.2 Many inputs: partial derivatives and the gradient

Terrain height depends on two inputs: position east (X) and north (Y). The *partial derivative* $\partial z / \partial X$ is the slope you feel walking purely east (holding Y fixed); $\partial z / \partial Y$ is the slope walking purely north. Pack the two into a vector, the *gradient*:

$$\nabla z = \left(\frac{\partial z}{\partial X}, \frac{\partial z}{\partial Y} \right). \quad (4.2)$$

The key idea

The gradient points in the direction of *steepest uphill*, and its length is how steep that is. On a hillside, ∇z is the arrow water would flow against. The overall steepness — the *slope* — is just the length of that arrow, $\|\nabla z\| = \sqrt{(\partial z / \partial X)^2 + (\partial z / \partial Y)^2}$, by Pythagoras.

So the slope angle of the ground is

$$\sigma_{\text{slope}} = \arctan \|\nabla z\| = \arctan \sqrt{(\partial_X z)^2 + (\partial_Y z)^2}, \quad (4.3)$$

combining the gradient (this chapter) with $\arctan$ (Chapter 2).

Where this lives in HAT1

Every roughness channel in HAT1 starts from this slope field. “Rough” ultimately means “the gradient is large and changes a lot from spot to spot.” The gradient is the first thing the pipeline computes from the raw elevation grid.

Practice

4.1 Find $f'(x)$ for $f(x) = 3x^2$ by the nudge method. **4.2** A patch has $\partial_X z = 0.1$ and $\partial_Y z = 0$ (metres per metre). What is the slope angle? Now $\partial_X z = \partial_Y z = 0.1$: is it steeper or shallower, and why does Pythagoras say so?

Chapter 5

Curvature and the Laplacian

The first derivative tells you the slope; the *second* derivative tells you how the slope is changing — whether the ground is bending into a bowl or over a dome.

For one input, the second derivative $f''(x)$ is just the derivative of f' . Positive f'' means the slope is increasing: the curve holds water (a bowl, like x^2). Negative f'' means a dome.

For terrain (two inputs) the natural single measure of bending adds the two straight-line curvatures:

$$\nabla^2 z = \frac{\partial^2 z}{\partial X^2} + \frac{\partial^2 z}{\partial Y^2} \quad (\text{the Laplacian}). \quad (5.1)$$

It is positive in a pit (a crater floor), negative on a peak (a rim or boulder top), and zero on anything flat or evenly tilted.

Intuition first

Stand a marble on the surface. Where it would settle (a pit), the Laplacian is positive; where it would roll off in every direction (a peak), negative; on a ramp, it neither settles nor has a preferred runaway, so zero. The Laplacian is a one-number “bowl-ness” meter — and a crater is a ring of negative rim around a positive floor, a signature the Laplacian sees clearly.

Where this lives in HATI

HATI’s curvature channel measures how much the Laplacian *varies* across a small window. Smooth ground has near-constant curvature; a crater field has curvature swinging from rim to floor. That swing is one of HATI’s strongest hazard signals.

Practice

5.1 For $f(x) = x^2$ we found $f'(x) = 2x$. What is $f''(x)$? Is it a bowl or a dome? **5.2** Sketch a cross-section of a crater (down then up) and mark where the curvature is positive and where negative.

Chapter 6

Pixels, finite differences, and convolution

Real elevation data is not a smooth formula — it is a *grid of numbers*, one height per pixel. So we need derivatives we can compute by arithmetic on neighbours.

6.1 Finite differences

The derivative is rise over run; on a grid the smallest run is one pixel. So the slope in the X direction at pixel (i, j) is approximately the right neighbour minus the left neighbour, over the distance:

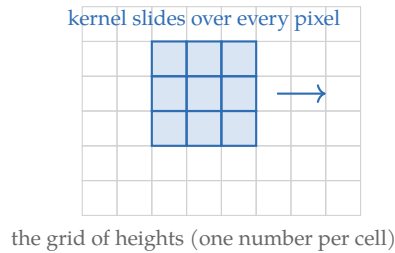
$$\frac{\partial z}{\partial X} \approx \frac{z_{i,j+1} - z_{i,j-1}}{2s}, \quad (6.1)$$

where s is the pixel size in metres. Curvature uses “neighbour + neighbour – twice the centre,” which on a grid becomes the five-point pattern $z_{i,j+1} + z_{i,j-1} + z_{i+1,j} + z_{i-1,j} - 4z_{i,j}$.

6.2 Convolution: sliding a little stencil

Doing the same little weighted combination of neighbours at *every* pixel has a name: *convolution*. You take a small grid of weights — a *kernel* — lay it over each pixel, multiply overlapping numbers, and add. The Sobel kernel for the east–west slope and the Laplacian kernel for curvature are

$$K_x = \begin{pmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{pmatrix}, \quad K_L = \begin{pmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{pmatrix}. \quad (6.2)$$



The key idea

A convolution is just “do the same small neighbour-recipe at every pixel.” Slopes, curvature, and even blurring are all convolutions with different little kernels. It is the workhorse that turns a static grid of heights into fields of slope and curvature.

Where this lives in HATI

The symbols $C_k[z]$ in the master equation are these operations: each channel is a convolution (or a close cousin) applied to the elevation grid z . The kernels above are literally the first lines of HATI’s code.

Practice

6.1 A row of heights is $\dots, 4, 4, 4, 9, 4, 4, \dots$ (one tall pixel). Apply the 1-D curvature recipe “left + right – 2 × centre” at the 9. Is the result positive or negative, and does that match “a bump”? **6.2** What does a kernel of all $\frac{1}{9}$ ’s (a 3×3 of ninths) do to an image? (Hint: it is an

average.)

Part III

Reading the data

Chapter 7

Describing a pile of numbers

A window of the slope field is just a pile of numbers. We need to summarise it.

Definition

For values $x_1, \dots, x_n$: the *mean* is $\bar{x} = \frac{1}{n} \sum x_k$ (the balance point). The *variance* is $\frac{1}{n} \sum (x_k - \bar{x})^2$ (average squared distance from the mean); its square root is the *standard deviation*, the typical spread. The *root-mean-square* (RMS) is $\sqrt{\frac{1}{n} \sum x_k^2}$ — the typical size, ignoring sign.

Those are *moment* summaries: they add things up. There is a second family that *sorts* instead. Line the values up smallest to largest. The *median* is the middle one. The value with a quarter of the data below it is the lower quartile Q_{25} ; three-quarters below, the upper quartile Q_{75} . Their gap is the *interquartile range*,

$$\text{IQR} = Q_{75} - Q_{25}, \quad (7.1)$$

a measure of spread that uses position, not arithmetic.

Intuition first

Mean and RMS *compute*; median and IQR *rank*. The difference matters because one wild value can drag a mean a long way but barely budes a median. We will lean on that robustness in the next chapter.

Where this lives in HATI

HATI's ten channels are exactly these summaries applied to slope and curvature inside a sliding window: RMS of slope, IQR of slope, IQR of curvature, and so on. "How rough is this patch" is answered by "how spread out are its slopes," and spread is mean-square or IQR.

Practice

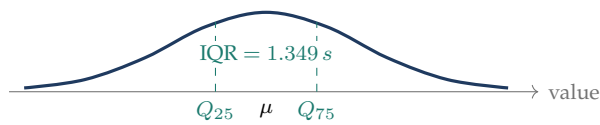
7.1 Find the mean and the median of $\{1, 2, 2, 3, 40\}$. Which one "feels" like the typical value, and why is the other misleading? **7.2** For $\{1, 2, 3, 4, 5, 6, 7, 8\}$ find Q_{25} , Q_{75} , and the IQR.

Chapter 8

The bell curve and robust statistics

8.1 The normal distribution

Measure many natural things and their histogram piles up in the famous symmetric “bell”: most values near a central one, fewer as you go out, tails thinning fast. This is the *normal* (or Gaussian) distribution, with a centre μ and a spread s . Its great convenience: fixed fractions of the data sit within fixed multiples of s from the centre, always.



8.2 Why median and IQR, and where 1.349 comes from

Mean and standard deviation are easy prey for outliers: a single huge value inflates both. Median and IQR shrug it off, because moving one extreme value does not move the middle or the quartiles. Statisticians call methods with this property *robust*.

But the IQR is on a different scale from the standard deviation. For a normal curve the quartiles sit at $\pm 0.6745 s$ from the centre (that 0.6745 is just where the bell cuts off a quarter of its area). So

$$\text{IQR} = 0.6745 s - (-0.6745 s) = 1.349 s \quad \implies \quad \hat{s} = \frac{\text{IQR}}{1.349}. \quad (8.1)$$

Dividing the IQR by 1.349 converts a robust spread back into ordinary standard-deviation units. That mysterious constant in the master equation is nothing but the width of a bell curve measured in IQRs.

Watch out

1.349 is *not* a tuning knob someone fiddled. It is forced by the geometry of the normal curve. If you ever see it and think “magic number,” remember: it is 2×0.6745 , and 0.6745 is the quartile of the standard bell.

Where this lives in HATI

The denominator $\text{IQR } g(C_k[z])/1.349$ in the master equation is this exact conversion, applied to each channel: it measures each roughness field’s spread robustly, then rescales it to standard-deviation units so all ten channels speak the same language.

Practice

8.1 A channel has $\text{IQR} = 2.698$. Estimate its standard deviation. **8.2** Add the value 1000 to the set $\{1, 2, 3, 4, 5\}$. By how much does the median move? The mean? Which is robust?

Chapter 9

Standardizing features

Ten channels in ten different units cannot be compared, let alone added. We put them on one ruler in three steps, all visible in the master equation.

1. **Tame the tail:** apply $g(x) = \log(1 + x)$. Roughness values have a few huge outliers; the log squashes them without changing their order.
2. **Centre and scale (the z -score):** subtract the median, divide by IQR / 1.349:

$$\zeta = \frac{g(C) - \text{med } g(C)}{\text{IQR } g(C) / 1.349}.$$

Now ζ is in “spreads from typical”: $\zeta = +2$ means “two spreads rougher than normal,” and it means the same thing in every channel.

3. **Clip:** cap ζ at ± 4 so no single freak pixel can dominate.

The key idea

Standardising strips away units and scale so different measurements become comparable. After it, “+2” is rough in every channel alike, and adding them up (the weighted sum) finally makes sense.

9.1 Ranking a result: percentiles

Once we have a hazard score for every pixel, we ask of one spot: *what fraction of the map is less hazardous than this?* That fraction is its *percentile*. The function that returns it — “share of values at or below t ” — is the *empirical cumulative distribution*. The Athena touchdown landed at the 96th percentile: rougher than 96% of the site.

Where this lives in HATI

The big bracketed fraction inside the master equation, log on top, IQR/1.349 on the bottom, with a clip _{$\pm c$} around it, is exactly this standardisation done to each C_k . You can now read the entire inside of the sum.

Practice

9.1 A channel value gives $g(C) = 3.0$; the field has median 1.0 and IQR = 1.349. Find ζ . **9.2** If a score beats 19 of the other 19 pixels in a 20-pixel scene, what percentile is it?

Part IV

The machine learning

Chapter 10

Probability in one bite

A *probability* is a number from 0 to 1 measuring how likely something is. We need only the simplest case: a yes/no outcome (*Bernoulli*). Label a site $y = 1$ if hazardous, $y = 0$ if safe. A model that outputs $H \in (0, 1)$ is claiming “Pr(hazard) = H .”

If the truth is $y = 1$ and the model said H , it assigned probability H to what happened; if $y = 0$, it assigned $1 - H$. A compact way to write “the probability the model gave to the actual outcome” is

$$H^y(1 - H)^{1-y}, \quad (10.1)$$

which equals H when $y = 1$ and $1 - H$ when $y = 0$ (anything to the power 0 is 1). This little expression is the seed of all the learning in the next chapter.

Intuition first

A good model gives high probability to things that actually happen. So if we collect the probabilities it assigned to every real outcome and multiply them, a better model makes that product bigger. “Make the product of assigned probabilities as large as possible” is, almost word for word, how machines learn.

Practice

10.1 A model says $H = 0.8$. Write the value of $H^y(1 - H)^{1-y}$ for the case $y = 1$ and for $y = 0$. Which outcome was it more confident about?

Chapter 11

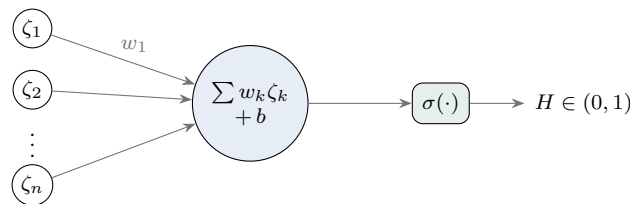
The logistic neuron

Now we assemble Parts I–III into the simplest machine-learning classifier, and discover it *is* HATI’s fusion step.

Take the feature vector ζ (the standardised channels), dot it with weights $\mathbf{w}$, add a bias b , and squash with the sigmoid:

$$H = \sigma(\mathbf{w} \cdot \zeta + b). \quad (11.1)$$

That is a *logistic-regression unit*, also called a single *neuron*. The weighted sum scores the evidence; the sigmoid turns the score into a probability.



The key idea

“Weighted sum, then sigmoid” — the skeleton from Chapter 1 — is a complete classifier. Stack many of these neurons and you have a neural network; the only new idea is how to *choose* the weights, which is the next chapter.

Where this lives in HATI

Equation (11.1) *is* the master equation, with $\mathbf{w} \cdot \zeta$ written as the dot product. HATI differs from textbook machine learning in one deliberate way: it does not *learn* $\mathbf{w}$ from data; it *sets* $\mathbf{w}$ from physics and the published literature. That is why it needs no training labels and why every weight has a stateable reason — the property that makes it trustworthy for a spacecraft.

Practice

11.1 With $\mathbf{w} = (1, 1)$, $b = -1$, and $\zeta = (2, 0)$, find H . Now $\zeta = (-2, 0)$. Interpret the two answers as “hazard” or “safe.”

Chapter 12

Learning the weights

HATI fixes its weights. But the natural next step — and the heart of machine learning — is to *learn* them from examples. Here is how, end to end.

12.1 A score for how wrong we are

From Chapter 10, a good model makes the product of assigned probabilities large. Multiplying many small numbers is awkward, so we take a logarithm (turning products into sums) and flip the sign (so “small is good”). The result is the *cross-entropy loss*:

$$J = -\frac{1}{D} \sum_{m=1}^D \left[y_m \log H_m + (1 - y_m) \log(1 - H_m) \right]. \quad (12.1)$$

J is large when the model confidently says the wrong thing, near 0 when it is confidently right. Learning = making J small.

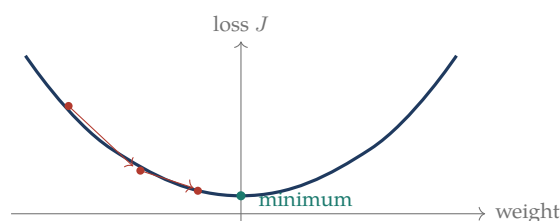
12.2 Rolling downhill: gradient descent

J depends on the weights. Picture J as a landscape and the weights as your position; you want the valley floor. The gradient $\partial J / \partial \mathbf{w}$ points uphill, so step the opposite way:

$$\mathbf{w} \leftarrow \mathbf{w} - \eta \frac{\partial J}{\partial \mathbf{w}}, \quad (12.2)$$

with a small step size η . Repeat until you stop descending. For this particular loss the gradient simplifies to something almost unbelievably clean:

$$\frac{\partial J}{\partial \mathbf{w}} = \frac{1}{D} \sum_{m=1}^D (H_m - y_m) \zeta_m. \quad (12.3)$$



Intuition first

Read Equation (12.3) in plain words: the correction to each weight is (*what the model predicted minus the truth*) times the feature. Predicted too little hazard on a rough feature? $H - y$ is negative, so the rule pushes that weight up. Too much? It pushes down. The model learns by repeatedly nudging weights in the direction that would have made its past mistakes smaller. Every neural network on Earth, however deep, is trained by versions of this one sentence.

Worked example

[one learning step, by hand] One sample, $\zeta = (2, 0)$, truth $y = 1$, current weights $\mathbf{w} = (0, 0)$, $b = 0$. Then $a = 0$, $H = \sigma(0) = 0.5$. The error is $H - y = -0.5$. The weight gradient is $(-0.5) \cdot (2, 0) = (-1, 0)$. Stepping with $\eta = 0.1$: $\mathbf{w} \leftarrow (0, 0) - 0.1 \cdot (-1, 0) = (0.1, 0)$. The first weight rose — the model nudged itself toward calling this rough sample hazardous. Repeat over many samples and the weights settle.

Where this lives in HAT1

HAT1 skips this loop on purpose, but the door is open: feed it labelled safe and hazardous sites and the same rule (12.3) would tune the ten weights. The reason HAT1 does not is strategic — fixed, physics-based weights need no training data (which barely exists for lunar hazards) and stay interpretable and certifiable. Knowing how learning *would* work is what lets you make that choice deliberately.

Practice

12.1 A sample has $\zeta = (1, -1)$, $y = 0$, and the model currently outputs $H = 0.7$. Find the error $H - y$ and the weight gradient $(H - y)\zeta$. Which weight goes up, which down, after a descent step? **12.2** Why do we take the log of the probabilities before summing, rather than multiplying them directly? (Think about $0.01 \times 0.01 \times \dots$)

The rest of machine learning, in a paragraph

Three ideas finish the toolkit, and you now have the footing for all of them. *Convexity*: for this loss the landscape is a single bowl (no false valleys), so descent always finds the true bottom — provable from the second derivative (the *Hessian*) being positive. *Regularization*: add a penalty $\frac{\lambda}{2} \|\mathbf{w}\|^2$ to keep weights modest and avoid memorising noise. *Backpropagation*: in a deep network, the same “error $\times$ input” rule is applied layer by layer, the error carried backward by the chain rule using the fact that $\sigma'(a) = \sigma(a)(1 - \sigma(a))$. That is the whole of how modern networks train; everything else is engineering.

Chapter 13

Why HATI is readable

A deep network gives an answer and no reason. HATI's single neuron gives both, and the maths of why is short. Because the score before the sigmoid is a plain sum,

$$a = b + \sum_{k=1}^{10} \phi_k, \quad \phi_k = \frac{w_k \zeta_k}{\|\mathbf{w}\|_1}, \quad (13.1)$$

each channel's *contribution* ϕ_k is a separate, addable number. So “why did this pixel score high?” has an exact answer: list the ϕ_k and see which were big. At the Athena touchdown, the slope-variation and small-crater-scale channels had large positive ϕ_k , while the average-tilt channels were near zero — the texture, not the steepness, raised the alarm.

The key idea

When the inside is a sum, attribution is exact and free: the whole equals the sum of its parts, so you can always say how much each part contributed. This is the deep reason HATI is auditable where a black-box model is not — and audibility is what a mission needs before it trusts a number.

Practice

14.1 Two channels have weights $w = (1, 0.6)$ (so $\|\mathbf{w}\|_1 = 1.6$) and standardised values $\zeta = (3, -1)$. Find each contribution ϕ_k and say which channel drove the score and in which direction.

Part V

Geometry and signals

Chapter 14

Matrices, eigenvectors, and the shadow ellipse

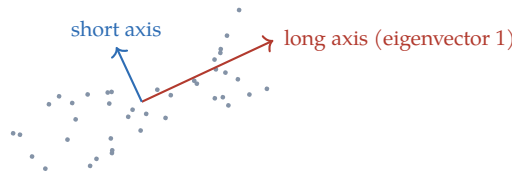
14.1 A matrix is a table that transforms vectors

A *matrix* is a rectangle of numbers. The only fact we need: a 2×2 matrix can stretch and rotate vectors. For special vectors — the *eigenvectors* — it only stretches, not rotates; the stretch factor is the *eigenvalue*. The eigenvectors are the natural axes of whatever the matrix describes.

14.2 Covariance: the shape of a cloud of points

Take the pixels of a shadow blob. Their spread has a shape — long one way, narrow the other. The *covariance matrix* captures it: the diagonal entries are how much the points spread east–west and north–south, the off-diagonal how much the two go together. Its eigenvectors point along the blob’s long and short axes; its eigenvalues are the squared lengths.

$$\mathbf{M} = \begin{pmatrix} \mu_{20} & \mu_{11} \\ \mu_{11} & \mu_{02} \end{pmatrix}, \quad \lambda_{1,2} = \frac{\mu_{20} + \mu_{02}}{2} \pm \sqrt{\left(\frac{\mu_{20} - \mu_{02}}{2}\right)^2 + \mu_{11}^2}. \quad (14.1)$$



Where this lives in HATI

HATI’s shadow census fits an ellipse to each dark blob with exactly this eigenvalue computation. The long axis (along the Sun) gives the shadow length L , hence the obstacle height via $h = L \tan e$ from Chapter 2; the short axis gives the footprint. The same algebra a statistician uses to find the axes of a data cloud finds the dimensions of a shadow.

Practice

15.1 A diagonal covariance $\mathbf{M} = \begin{pmatrix} 9 & 0 \\ 0 & 1 \end{pmatrix}$ has eigenvalues 9 and 1 along the axes. Which way is the blob long, and by what ratio of lengths (recall length $\propto \sqrt{\lambda}$)?

Chapter 15

Mapping a sphere onto a sheet

The Moon is a ball; our map is flat. The bridge is the *stereographic projection*, and it comes from one picture and similar triangles.

Stand at the north pole N . For any surface point P , draw the straight line from N through P and continue it until it strikes the flat sheet laid against the south pole. Where it strikes is where P goes on the map. Two similar triangles in the slice through N , P , and the pole give the map distance from the centre as

$$\rho = 2R \tan\left(\frac{\theta}{2}\right), \quad (15.1)$$

where R is the Moon's radius and θ the angle of P from the south pole. Writing θ in terms of latitude φ and adding the longitude λ for direction gives the working formula

$$\rho = 2R \tan\left(\frac{\pi}{4} + \frac{\varphi}{2}\right), \quad X = \rho \sin \lambda, \quad Y = \rho \cos \lambda. \quad (15.2)$$

Intuition first

Shine every south-polar point up to the north pole and catch its shadow on a sheet under the Moon. Near the south pole this barely distorts anything — a metre on the map is a metre on the Moon — which is exactly the regime a polar landing cares about.

Where this lives in HATI

This is how HATI turns the published touchdown latitude and longitude into a specific row and column on the elevation grid, so it can read the hazard score at the exact spot the lander came down. Geometry, not guesswork, places the pixel.

Practice

16.1 At the south pole $\varphi = -90^\circ$. Show $\rho = 0$, i.e. the pole maps to the centre. (Use $\tan(\frac{\pi}{4} + \frac{\varphi}{2})$ with $\varphi = -\frac{\pi}{2}$.)

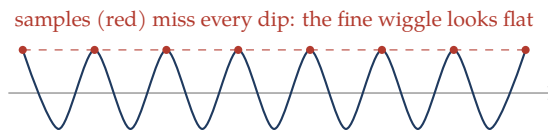
Chapter 16

Sampling and the resolution floor

Why is a 4 m map “blind below 8 m”? Because of how sampling works — the same reason a low-quality audio recording loses the high notes.

When you record a signal by taking a value every s metres (or every fraction of a second), you can only faithfully capture wiggles slower than *two samples per wiggle*. Anything finer — a bump that rises and falls within a single pixel — cannot be distinguished from something smoother; it “folds” into a false, slower shape. This is the *Nyquist–Shannon sampling theorem*, and the shortest wavelength you can trust is twice the sample spacing:

$$\lambda_{\min} = 2s. \quad (16.1)$$



For HATI’s $s = 4$ m elevation map, $\lambda_{\min} = 8$ m: craters and rocks smaller than that simply are not in the data. They are not faint — they are absent.

Where this lives in HATI

This floor is the entire reason HATI has a second channel. The shadow census runs on a 0.9 m photograph, whose floor is far lower, so it sees the sub-8 m hazards the elevation map physically cannot. The two channels are not redundant; they cover different size bands by design.

Practice

17.1 A different DEM has 1.5 m pixels. What is its resolution floor? **17.2** In one sentence, why can a shadow in a 0.9 m photo reveal a 2 m rock that a 4 m elevation map cannot?

Part VI

The summit

Chapter 17

Reading the master equation

You now own every piece. Let us read the wall, left to right, the way you would read a sentence.

$$H(x) = \sigma \left(b + \frac{1}{\|\mathbf{w}\|_1} \sum_{k=1}^{10} w_k \operatorname{clip}_{\pm c} \left[\frac{g(C_k[z])(x) - \operatorname{med} g(C_k[z])}{\operatorname{IQR} g(C_k[z])/1.349} \right] \right).$$

- z is the elevation grid; $C_k[z]$ are the ten roughness fields, each a **convolution / windowed statistic** (Chapters 6–7) of slope and curvature (Chapters 4–5) computed from the grid.
- $g = \log(1 + \cdot)$ **tames the tail** (Chapter 1); subtracting the median and dividing by $\operatorname{IQR} / 1.349$ is the **robust z -score** (Chapters 8–9) that puts all ten channels on one ruler; $\operatorname{clip}_{\pm c}$ caps outliers.
- $\sum_k w_k(\dots)$ is the **weighted sum / dot product** (Chapters 1, 3): each standardised channel times its importance, added. Dividing by $\|\mathbf{w}\|_1$ keeps the scale steady.
- b is the bias; $\sigma(\dots)$ is the **logistic squash** (Chapter 1) that turns the score into a hazard probability $H(x) \in (0, 1)$ — the **logistic neuron** of Chapter 11, with weights set from physics rather than learned (Chapter 12), and **exactly attributable** channel by channel (Chapter 14).

The key idea

Read aloud: “Take the elevation grid; measure ten flavours of roughness; put each on a common, outlier-resistant scale; add them up with physics-given importances; squash the total into a probability.” That sentence is `HATI`. You can now say it, and defend every word.

Alongside it runs Channel 2: detect shadows in the fine photograph, fit each an ellipse by the covariance eigenvalues (Chapter 15), read its length and footprint, convert length to height with $h = L \tan e$ (Chapter 2), and so census the obstacles that live below the elevation map’s 8 m floor (Chapter 17). The stereographic projection (Chapter 16) pins both channels to the right pixel.

So how do you become an expert?

Understanding is the door; expertise is the room. Three steps through it. First, do every `PRACTICE` box with a pen — the maths becomes yours only when your own hand makes it. Second, read `hati_math.pdf` again: the formal document that this book was the key to; it will now read as plain prose. Third, change something — recompute a channel, alter a weight, run the pipeline on a new patch, and predict the effect before you look. The moment you can predict what the equation will do before the computer tells you, you are no longer learning the topic. You are doing it.

Chapter A

Notation at a glance

z, z_{ij}	elevation grid and the height at row i , column j
s	pixel size in metres (the ground sample distance)
∇z	gradient: the steepest-uphill arrow; its length is the slope
$\nabla^2 z$	Laplacian: curvature; + in a pit, – on a peak
$C_k[z]$	the k -th roughness channel (a convolution / windowed statistic)
$g(x) = \log(1 + x)$	the tail-taming log
med, IQR	median and interquartile range (robust spread)
ζ_k	the k -th channel after standardising (a z -score)
$\mathbf{w}, b$	the weight vector and bias
$\ \mathbf{w}\ _1 = \sum_k w_k $	the ℓ_1 norm (sum of weight sizes)
$\sigma(a) = 1/(1 + e^{-a})$	the logistic / sigmoid
$H(x)$	the hazard probability at spot x , between 0 and 1
$h = L \tan e$	obstacle height from shadow length L and Sun elevation e
$\lambda_{\min} = 2s$	the resolution floor (Nyquist)
$\phi_k = w_k \zeta_k / \ \mathbf{w}\ _1$	channel k 's contribution to the score

Chapter B

Solutions to the practice boxes

- 1.1** $\sigma(0) = \frac{1}{2}$; $\sigma(2)$ is closer to 1 ($a > 0$). **1.2** $a = 2(1.5) - 1(1) - 1 = 1$; $\sigma(1) \approx 0.73$. **1.3** $\log 0 = -\infty$, undefined for a zero-valued channel; $\log(1 + 0) = 0$ is fine.
- 2.1** $h = 6 \tan 20^\circ \approx 2.2$ m. **2.2** $L = h / \tan 5^\circ \approx 2.2 / 0.0875 \approx 25$ m; a low Sun reveals small hazards.
- 3.1** $1(2) + 0.7(-1) + 0.7(3) = 2 - 0.7 + 2.1 = 3.4$. **3.2** No — positive weights mean a larger input only raises the sum; one rough channel can only push the hazard vote up.
- 4.1** $f'(x) = 6x$. **4.2** $\arctan(0.1) \approx 5.7^\circ$; with both 0.1, $\|\nabla z\| = \sqrt{0.02} \approx 0.141$, angle $\approx 8.0^\circ$ — steeper, because the two slopes combine by Pythagoras.
- 5.1** $f''(x) = 2 > 0$: a bowl. **5.2** Curvature is positive on the floor, negative at the rims.
- 6.1** $4 + 4 - 2(9) = -10 < 0$: a peak (the recipe is the Laplacian; a bump gives negative). **6.2** It blurs (replaces each pixel by the average of its 3×3 neighbourhood).
- 7.1** mean = 9.6, median = 2; the median is typical, the mean is dragged by 40. **7.2** $Q_{25} = 2.5$, $Q_{75} = 6.5$, IQR = 4.
- 8.1** $\hat{s} = 2.698 / 1.349 = 2$. **8.2** median moves from 3 to 3.5 (barely); mean jumps from 3 to 169.2 — the median is robust.
- 9.1** $\zeta = (3 - 1) / (1.349 / 1.349) = 2$. **9.2** 19/19 below it among the others $\Rightarrow$ the top, the 100th (or, of all 20, the 95th) percentile.
- 10.1** $y = 1$: 0.8; $y = 0$: 0.2; more confident about hazard.
- 11.1** $(2, 0)$: $a = 1$, $H \approx 0.73$ (hazard); $(-2, 0)$: $a = -3$, $H \approx 0.05$ (safe).
- 12.1** error 0.7; gradient $0.7(1, -1) = (0.7, -0.7)$; after $-\eta$ grad, w_1 down, w_2 up. **12.2** multiplying many small probabilities underflows to 0; logs turn the product into a stable sum.
- 14.1** $\phi_1 = 1 \cdot 3 / 1.6 = 1.875$, $\phi_2 = 0.6 \cdot (-1) / 1.6 = -0.375$; channel 1 drove it strongly upward.
- 15.1** long along the eigenvalue-9 axis; length ratio $\sqrt{9} : \sqrt{1} = 3 : 1$.
- 16.1** $\tan(\frac{\pi}{4} - \frac{\pi}{4}) = \tan 0 = 0 \Rightarrow \rho = 0$.
- 17.1** $2 \times 1.5 = 3$ m. **17.2** the rock's shadow spans several 0.9 m pixels even though the rock is smaller than one 4 m elevation pixel.

Chapter C

Where to go next

- `hati_math.pdf` — the rigorous companion this book unlocks.
- Calculus and linear algebra: any first-year university text, or the free *3Blue1Brown* “Essence of” video series for the pictures behind Chapters 4–5 and 15.
- Machine learning: the logistic-regression chapter of any introductory course gives Chapters 11–13 in full.
- Lunar terrain: Kreslavsky & Head (2000) and Rosenberg et al. (2011) for the roughness operators; Hapke (1984) for the shadow photometry; Henriksen et al. (2017) for how the elevation maps are made.